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Compensating Thermal Derated Torque of IPMSM Centric Electric Vehicles

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ABSTRACT A novel control approach based on Linear Parameter Varying (LPV) is designed for Electric Vehicles (EVs) to address the crucial challenge of torque derating. A high fidelity Interior Permanent Magnet Synchronous Motor (IPMSM) based EV is modelled by taking into account the influence of temperature. A closed loop torque compensation control architecture with LPV based Field Oriented Control (FOC) has been designed to mitigate the thermal effects. In order to curtail deteriorating of FOC due to parameter variation, a robust LPV observer has been proposed to estimate the degraded Permanent Magnet (PM) flux linkage and instantaneous torque production caused by uncertainties in stator resistance arises due to thermal effects. A LPV gain scheduling robust current controller for inner control loop of FOC has also been designed which ensures internal stability. The LPV observer-controller gains are computed by solving Linear Matrix Inequalities (LMIs) using convex optimization and singular value decomposition. The experiments simulations have been performed for EV operation against the Worldwide Harmonized Light Vehicles Test Procedure (WLTP) class 3, New European Driving Cycle (NEDC) and Federal Test Procedure (FTP) driving cycles considering thermal effects in Matlab/Simulink. The efficacy of proposed control architecture for managing the derating torque of IPMSM based EV is demonstrated through the validity of results and detailed data analysis. The LPV based FOC is compared with conventional FOC control counterpart. The comparative experiments simulation results show that the overall Mean Square Error (MSE) value is exponentially reduced which establish superiority of the proposed control scheme.

INDEX TERMS Derated torque, electric vehicles (EVs), linear parameter varying (LPV) control, interior permanent magnet synchronous motor (IPMSM), field oriented control (FOC).

NOMENCLATURE

τ_L	Load torque.	ψ_T, ψ_{T_0}	PM flux linkage at operating and nominal temperatures T .
$v_{cycle}(t)$	Desired Vehicle Speed.	τ_{eRef}	Reference torque for operating period.
α	Temperature coefficient of remanence.	θ_m	Rotor position.
β	Temperature coefficient of coercivity.	B	Viscous Damping coefficient.
$\Delta(T)$	Difference between operating and nominal temperature.	B_r, B_{r0}	Magnet remanence at operating and nominal temperature.
$\hat{\psi}_d, \hat{\psi}_q$	Estimated flux of in d and q axis.	F_a	Aerodynamics drag force.
$\hat{\psi}_{PM}$	Estimated PM flux linkage.	F_r, F_t	Rolling and downgrade resistance force.
ω_m	Rotor mechanical speed.	F_{te}	Tractive force.
ψ_d, ψ_q	Flux in d and q axis.	F_v	Viscous frictional force.
ψ_{PM}	PM flux linkage.	H_c, H_{c0}	Magnet coercivity at operating and nominal temperature.
The associate editor coordinating the review of this manuscript and approving it for publication was R. K. Saket ¹ .		i_d, i_q	Stator currents in d and q axis.
		J	Rotor Inertia.

k_d	Coefficients for Derivative values.
k_p, k_i	Coefficients for Proportional and Integral values.
L_d, L_q	Inductances in d and q axis.
p	Number of poles.
R_{sT}, R_{s0}	Winding resistance at operating and nominal temperatures.
V_d, V_q	Stator voltages in d and q axis.

I. INTRODUCTION

Legislation for reduction in emissions, rapid evanescence and technological aspects have made the automotive industry go under radical transformation from Internal Combustion Engine (ICE) towards the electrification of the transportation sector. EVs categorized as Fully Electric Vehicles (FEVs), Plug-in Hybrid Electric Vehicles (PHEVs) and Hybrid Electric Vehicles (HEVs) are being acknowledged as potential sustainable options to combat economic growth and concerns over environmental situation [1], [2]. The increased complexity of the electrified powertrain associated with these EVs offer interesting control challenges for electric traction motors being the pivotal component: Robust performance in different operating environments, fast and abrupt torque response, constant torque at vast speed range, minimum torque ripples, high torque, lower amount of inertia, small size and reduced weight [3]. Keeping in view the required traction motor characteristics, recent studies [4], [5] suggested Permanent Magnet Synchronous Motor (PMSM) will be privileged in near future among Induction Motor (IM), DC motor, and Switched Reluctance Motor (SRM). In further classification PMSM has two types: IPMSM and Surface Permanent Magnet Mounted Synchronous Motor (SPMSM). IPMSM one of the radial flux types in which permanent magnets are buried inside the rotor and inductance of q-axis larger than the d-axis is commonly used in automotive industry because of its mechanical strength for high speed applications. On the other hand, in SPMSM permanent magnets are mounted on the rotor and has equal inductance in d-q axis.

The IPMSM has been widely used in electrified power train by Toyota (Rav4 and prius), Volkswagen (ID.3), Fiat (Peugeot ION), Honda, Hyundai (Blueon), Nissan (leaf and Altera), Chevrolet (Volt), Tata (Indica), BYD (F3M and e6), and Mitsubishi (Miev) [6], [7]. The PMSM machine is not limited to automotive industry but has popularly emerged in aerospace and marine industries as well.

For efficient performance of IPMSM drive, being the most significant part of EV, many distinguished control strategies have been demonstrated including sliding mode controls catering the chattering effects [8], [9], intelligent control methods [10], [11], and modern control strategies [12], [13]. However, FOC and Direct Torque Control (DTC) are the most common and effective methods adopted for speed and torque control [14], [15]. Even though DTC is considered due to less dependency of parameters but does not perform well at low motor speeds because of high torque ripples.

FOC is extensively used for high performance traction applications due to its high reliability, simple algorithm, strong robustness, and wide speed range with high torque at zero speed, which are the utmost desired characteristics of traction motors [16]. Conventional FOC compromises performance and fails to sustain the desired closed loop stability in high performance traction applications. To overcome these shortcomings, recently, LPV (gain scheduling) control has attracted the researchers due to explicit exploitation of knowledge of real time dynamics of a traction motor [17], [18]. The innovative idea was first introduced [19] which provided systematic, efficient control approach while ensuring closed loop stability and performance.

Moreover, FOC is also sensitive to parameter variations. In IPMSM parameters are affected due to change in operating environment caused by drive cycles (highway and urban traffic conditions), vehicle load and temperature influence [20], [21]. Consequently, effecting the flux and torque production, resulting in thermal derating of the electrified powertrain. Therefore, for effective implementation, FOC requires precise and accurate knowledge of thermally degraded flux and hence, the torque production.

In order to estimate the torque considering magnetic flux linkage as a function of temperature, already proposed approaches are roughly classified into indirect and direct methods [22], [23]. The indirect methods mainly include: the torque estimation through reactive power feedback where singularity is an issue, Disturbance Input Decoupling (DID) not accurate at low speed, perturbation searching not guaranteeing the dynamics performance and invasive method of high frequency current signal injection producing additional losses in real systems. The direct methods consist of infrared thermography which is limited to the surface of rotor magnet. It is also not a viable solution due to high cost and construction drawbacks depending on magnets and winding temperature [24]. Moreover, the extra sensors do not work during motor operation for thermal modelling [25].

In recent past various online estimation techniques in the literature are also presented [26], [27], to increase the robustness of control strategy which can be classified as: Numerical Methods (Recursive least square [28], affine projection algorithm [29], Extended Kalman Filtering (EKF), Kalman like adaptive observers [30] and model reference adaptive system [31]) suffer from high and complex computations. Artificial Intelligence (AI)-based methods [32] and particle swarm optimization methods [33] require large dataset and hence very high computation cost and long time for training. Observe based methods including disturbance observers which are robust but require accurate dynamics of nonlinear model [34], luenberger and extended luenberger method [35], and adaptive flux observers are also employed [36]. Due to robustness, outstanding transient behavior and good state estimation, Sliding Mode Observer (SMO) technique is into practice for long time [37]. SMO based FOC is also employed for PMSM drive using enhanced linear extended state observer for rotor position mitigating chattering effects

TABLE 1. Comparison with other FOC techniques.

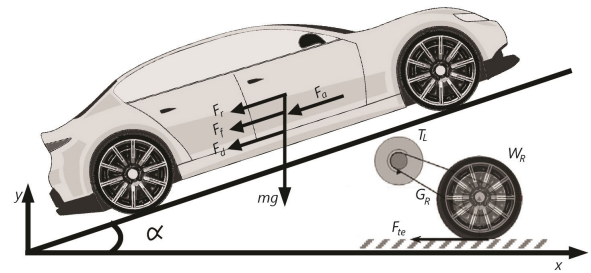
Existing Methodologies	ERROR (%)	
	Referred	Proposed LPV based FOC
LPV Control [39]	1.1	0.014
LPV H_∞ Control [40]	0.425	0
AI Control [32]	23.1	23.5
FOSMC Control [8]	3.2	0.15

but increases computational burden which makes it difficult for practical implementation [38].

Linear Parameter Varying (LPV) is a potential solution to overcome these control challenges. In order to validate the robustness and effective of proposed LPV based FOC, comparative analysis with FOC based techniques are illustrated in Table 1. To compare the presented technique, similar reference profile reported in publications are generated and percentage of error in terms of the performance indice presented in the publications are depicted in the Table 1. In [39] LPV gain scheduled output feedback controller without observer has been proposed for speed control of SPMSM. The SPMSM is not a preferred choice for traction application due to its mechanical structure. The LPV controller does not consider temperature effects for derating of torque. H_∞ LPV control with state estimator and a nonlinear torque modulation has been designed for PMSM speed problem [40]. The technique does not takes into account temperature effects. Both these techniques determined good performance when compared with presented proposed scheme and easy deployability properties. AI based [32] also performs well, but it is considered an expensive solution due to high computation cost and long training time. The percentage error of Field Oriented Sliding Mode Control (FOSMC) is compared which too is very high in comparison to presented work. The comparative analysis in terms of the percentage of error established the superiority of the proposed control scheme.

To the best of author's knowledge, estimation of flux and torque due to change of operating temperature with LPV technique based on FOC to cater the derating torque of IPMSM in electrified powertrain has not previously been reported. Motivated by the importance of aforementioned observations, the major contributions and proposed control strategy are as follows:

- Derivation of nonlinear IPMSM dynamic model considering thermal effects and emulation in LPV form.
- Design of LPV based FOC control strategy for compensation of thermally derated torque of IPMSM taking into account EV aerodynamics and mechanics. The FOC outer loop generates the desired currents from PM flux and torque controllers employed using Proportional Integral Derivative (PID) control. For estimation of flux and torque due to change in operating temperature, LPV observer is constructed using LMI.

**FIGURE 1.** Forces acting on electric vehicles.

- The inner loop of LPV based FOC is designed using LPV gain scheduling robust current feedback controller, scheduled with varying stator resistance and rotor speed.
- The performance of novel observer-controller technique is evaluated considering WLTP class 3, NEDC and FTP standard driving cycles commonly adopted by the automotive community [5]. The results and detail data analysis in terms MSE and percentage error are able to demonstrate effectiveness of the control strategy for compensating the derating of torque for wide operating conditions.
- The proposed control scheme is also compared with conventional FOC. Comparative analysis of the experiments result establish the efficacy and robustness of LPV based FOC.

The remainder of this article is structured in six sections: Section II presents a brief introduction to the proposed control framework. Section III presents EV dynamics including IPMSM mathematical modeling taking into account the temperature effects, reformulation to LPV model and elaboration of performance criterion; section IV discusses control consisting of novel robust observer and LPV gain scheduling technique for current controller; section V shows EV performance evaluation results including testing of control strategy based on standard driving cycles; section VI presents the comparative analysis of the FOC based LPV observer-controller pair with conventional FOC followed by conclusion in section VII.

II. PROPOSED LPV BASED FOC CONTROL ARCHITECTURE FOR IPMSM CENTRIC ELECTRIFIED POWERTRAIN

This section holistically presents the proposed LPV based FOC architecture for IPMSM traction motor commonly used in EVs. The Electrified Powertrain consisting of proposed LPV based FOC control framework, IPMSM and gearbox with torque inputs and outputs, and the rest of vehicle dynamics as shown in Fig. 2. In real time conditions, IPMSM in electrified powertrain is controlled to deliver the required torque against the road loads using LPV based FOC control strategy. EV mechanics in section III is used to transmit IPMSM speed generated in (rpm) through gearbox to EV speed in (km/h) and torque against the driving cycles simulated in V-A. The novel designed torque estimator in IV-A and reference torque error is then fed into torque controller

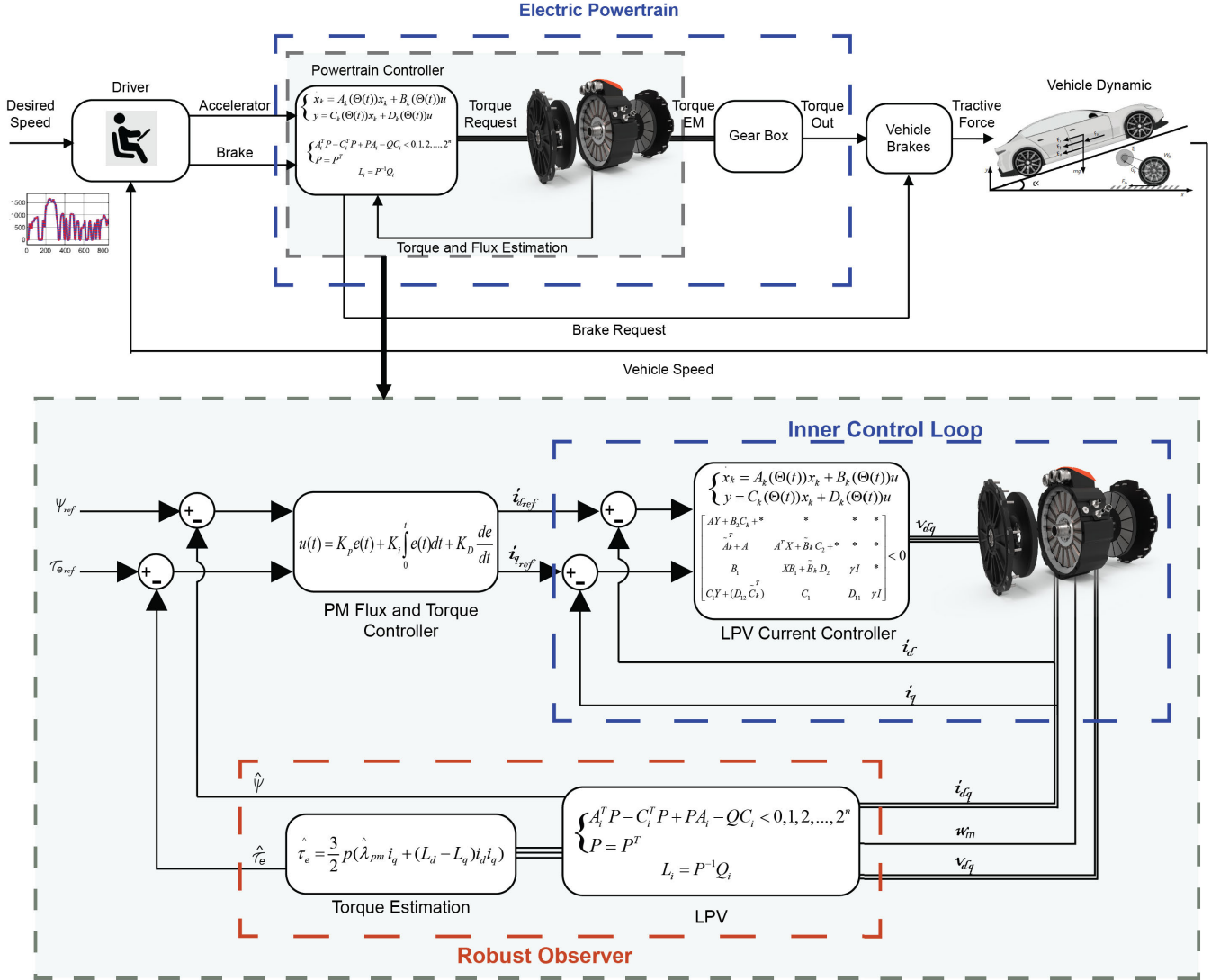


FIGURE 2. Proposed control architecture for IPMSM centric electric vehicle.

designed in section IV-C. The error between measured currents and from torque controller is used by LPV current controller designed in section IV-B. The control architecture is also fed with measured and estimated speed, voltages, and flux. The design details have been discussed in the subsequent sections.

III. EV AND IPMSM DYNAMICS

This section discusses the EV mechanics in section III-A taking into consideration the non linear IPMSM mathematical model with effects of temperature stated in section III-B and III-C. To implement the LPV based FOC gain scheduling controller the IPMSM model needs to be reformulated which is presented in section III-D. For evaluation of the control architecture, performance criterion has been mentioned in section III-E.

A. EV MECHANICS

The demonstrated control strategy in section IV. is not solely applied to IPMSM but takes into account the EV aerodynamics and mechanics principles depicted in Fig.(2) [41]. The speed dynamics of wheel drive and load torque in equation (1 and 2) described in the traction motor referential is as follows:

$$\dot{\omega}_m = \frac{p}{J} (\tau_e - \tau_L - B\omega_m) \quad (1)$$

$$\tau_L = (F_r + F_f + F_a + F_d)w_r \quad (2)$$

Furthermore, IPMSM speed is proportional to vehicle speed (v), which can be expressed in terms of gear box ratio (G_r) and wheel radius (w_r) by:

$$\omega_m = \frac{G_r}{w_r} v \quad (3)$$

B. NONLINEAR IPMSM MODEL

The three phase nonlinear IPMSM model after applying Clarke and Park transform can be represented in d-q coordinates from equation (4) to (10). The mathematical model being the most exclusive [42] after taking into consideration different models in research within [42]–[46]. In order to reduce down to essential; saturation, coupling, spatial harmonics and core losses are assumed negligible of nonlinear mathematical model.

$$V_d = R_s i_d + \frac{d}{dt} \psi_d - p \omega_m \psi_q \quad (4)$$

$$V_q = R_s i_q + \frac{d}{dt} \psi_q + p \omega_m \psi_d \quad (5)$$

$$\begin{cases} \psi_d = L_d i_d + \psi_{PM} \\ \psi_q = L_q i_q \end{cases}$$

The electromagnetic torque equation is as:

$$\tau_e = \frac{3}{2} p (i_q \psi_{PM} + (L_d - L_q) i_d i_q) \quad (6)$$

The IPMSM model can be rewritten in explicit form by opting the state variables $(\theta_m, \omega_m, i_d, i_q)$ as follows:

$$\dot{\theta}_m = \omega_m \quad (7)$$

$$\dot{\omega}_m = \frac{3p}{2J} [i_q \psi_{PM} + (L_d - L_q) i_d i_q] - \frac{B}{J} \omega_m - \frac{1}{J} \tau_L \quad (8)$$

$$\dot{i}_d = \frac{1}{L_d} (-R_s i_d + p L_q \omega_m i_q + V_d) \quad (9)$$

$$\dot{i}_q = \frac{1}{L_q} (-R_s i_q + p L_d \omega_m i_d - p \psi_{PM} \omega_m + V_q) \quad (10)$$

C. TEMPERATURE EFFECTS ON IPMSM

To determine performance of traction motor, thermal analysis plays a vital role. In electrified powertrain of EVs, IPMSM normally faces extreme environmental conditions resulting in temperature variation which effects the core material in equations (11 and 12), inductance, stator resistance, PM flux and hence torque production. PM flux linkage plays significant role in derating the torque production accuracy and the efficiency of traction motor. The temperature influenced flux effects the torque directly, as the temperature rises, permanent magnet flux decreases and hence the torque production decreases linearly [21], [47], [48]. The equation (13) presents the temperature influenced torque [21]. It is worth mentioning that temperature coefficient of remanence and coercivity α and β are not constant instead varies with temperature.

$$B_r = B_{r0} [1 + \alpha \frac{\Delta(T)}{100}] \quad (11)$$

$$H_c = H_{c0} [1 + \beta \frac{\Delta(T)}{100}] \quad (12)$$

$$\tau_e(T) = \frac{3}{2} p [\psi_{T0} i_q + (L_{sd} - L_q) i_d i_q] + (\alpha \frac{T - T_0}{100} \psi_{T0} i_q) \quad (13)$$

Moreover, in this paper changes due to variation in temperature including reduction of losses in core material resulting

in decrease conductivity is also assumed negligible because it happens typically over 200°C (not applicable in traction motor) [49], [50]. The stator resistance can be represented in equation (14) where $\gamma_0 = 3.93 \times 10^{-3}/^\circ\text{C}$ temperature coefficient of copper at nominal temperature of 20°C .

$$R_{sT} = R_{s0} (1 + \gamma_0 \Delta(T)) \quad (14)$$

The produced temperature is based on the effects of temperature on the flux linkage of IPMSM, hence torque production [51].

D. LPV MODEL FORMULATION

The mathematical model of IPMSM needs to be reformulated in order to implement the proposed LPV gain scheduling control technique. The nonlinear model of IPMSM defined in (7-10) can be rewritten in LPV system described by the following state space equation:

$$(G(\Theta), P_\Theta) = \begin{cases} \dot{x} = A(\Theta(t))x + B(\Theta(t))u \\ y = C(\Theta(t))x + D(\Theta(t))u \end{cases} \quad (15)$$

The parameter dependent state space matrices A, B, C, and D are continuous function of scheduling variable $[\Theta] = [w_m \ R_s]^T$ and are considered as internal variables with state vector, control inputs and output; $x = [i_d \ i_q]^T$, $u = [v_d \ v_q]^T$, and $y = [i_d \ i_q]^T$.

In LPV model of IPMSM, P_Θ is a convex polytopes with vertices, Θ_i , $i = 1, 2, \dots, N$. and can be defined as:

$$P_\Theta = \text{Co}\{\Theta_{vi}, \Theta_{vi}\} \quad (16)$$

where, Θ_{vi} are the vertices, $i = 1, 2, \dots, L$. $L = 2^l$ are the number of vertices. Co is a convex hull, i.e. the set of all convex combinations of Θ_{vi} (all points inside and on the boundary of the polytopes). The matrix $A(\Theta(t))$ can be given as:

$$\begin{aligned} A(\Theta(t)) &= A_0 + A_1 \Theta(1) + A_2 \Theta(2) \\ A(\Theta) &= \begin{bmatrix} -R_s/L_d & L_q p \omega_m / L_d & 0 & 0 \\ 0 & -R_s/L_q & -p \omega_m / L_q & 0 \\ -R_s & 0 & 0 & p \omega_m \\ 0 & -R_s & -p \omega_m & 0 \end{bmatrix} \\ B(\Theta) &= \begin{bmatrix} 1/L_d & 0 \\ 0 & 1/L_q \end{bmatrix} \end{aligned} \quad (17)$$

$C = [I]$, $D = [O]$, $I = 2 \times 2$ identity matrix and $O = 2 \times 4$ zero matrix.

Thus, with the definition of varying parameters, the nonlinear IPMSM dynamics can be implemented as LPV model for gain scheduling current controller in section IV-B.

E. EV PERFORMANCE CRITERION

The performance of EV is substantially based on the speed and torque production which are influenced by operating temperature. For smooth functioning of EV, following norms are considered:

$$\min_{I_d, I_q, w_m} \|\tau_e(t) - \tau_{eRef}(t)\| \quad (18)$$

$$\min_{I_d, I_q, w_m} \|v(t) - v_{cycle}(t)\| \quad (19)$$

IV. CONTROL ARCHITECTURE DESIGN

A comprehensive architecture of IPMSM control structure with FOC based inner and outer loop shown in zoomed part of Fig. (1), entailed by four different exclusively designed modules. The LPV current controller block acts as stator current feedback controller scheduled with varying stator resistance R_s and rotor speed ω_m . The robust current output feedback controller ensures good tracking over the entire operating range. A novel robust LPV observer depicted in section IV-A is designed to estimate the thermal effects, the LPV gain scheduling current controller has been designed in section IV-B. Finally, torque and flux controller is designed using PID explained in section IV-C. Altogether, these controllers and observer ensure management of the thermal derating of torque. To verify the aforementioned observer-controller pair WLTP class 3, NEDC and FTP driving cycles are used to generate reference speed and load torque required as represented in Fig. 4 (a, b, c and d).

A. LPV OBSERVER

Load torque of EV due to aerodynamics drag, rolling resistance and road grade force changes, depicted in equation (2). The increase of torque propel EV to move forward with a speed mentioned in equation (3). The speed and torque of EV connected through gearbox to IPMSM is of great importance in performance degradation. It causes flux variation and derating torque which need to be estimated. The accurate flux estimation is necessary not only for the optimal control but also to avoid permanent damages. Robust LPV observer is designed for flux estimation and hence torque which guarantees exact convergence. LPV observer is designed in stationary reference frame in equations (20-23) to cater thermal effects due to variation of stator resistance given in equation (14).

$$\begin{bmatrix} \dot{\hat{\psi}}_d \\ \dot{\hat{\psi}}_q \end{bmatrix} = A(\Theta) \begin{bmatrix} \hat{\psi}_d \\ \hat{\psi}_q \end{bmatrix} + B(\Theta)u + L(\Theta)C(\Theta) \left(\begin{bmatrix} \psi_d \\ \psi_q \end{bmatrix} - \begin{bmatrix} \hat{\psi}_d \\ \hat{\psi}_q \end{bmatrix} \right) \quad (20)$$

The observer gain L is computed by solving following LMIs formulated as follows:

$$\begin{cases} A_i^T P - C_i^T Q_i^T + P A_i - Q_i C_i < 0, & i = 1, \dots, 2^\Theta \\ P = P^T \end{cases} \quad (21)$$

The following equation gives the gain of local observer:

$$L_i = P^{-1} Q_i \quad (22)$$

The linear combination of local observer in equation (22) computes the LPV observer as follows:

$$L(\Theta) = \sum_{i=1}^{2^\Theta} c_i L_i \quad (23)$$

The estimated torque is given by:

$$\hat{\tau}_e = \frac{3}{2} p (i_q \hat{\psi}_{PM} + (L_d - L_q) i_d i_q) \quad (24)$$

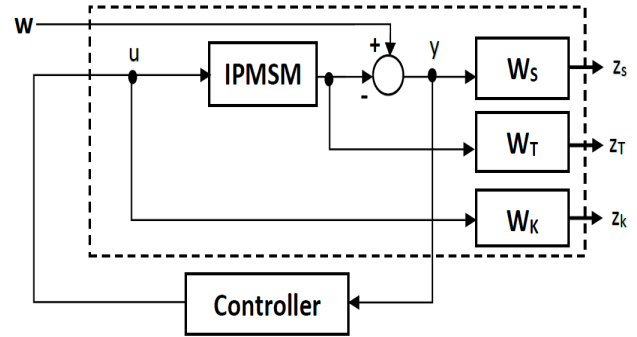


FIGURE 3. Generalized plant with weights.

The design procedure for LPV observer is outlined below:

- IPMSM is modeled into affine LPV form by modeling rotor speed and stator resistance formulated in equation (17).
- For computing the LPV observer gain, the observer error is defined in equation (20).
- Quadratic stability is ensured and LMI in equation (21) is solved for local observer gain in equation (22).
- The linear combination of local observer computes the LPV observer gains mentioned in equation (23).

B. LPV CURRENT CONTROL DESIGN

The estimated torque from the previous section is then fed into designed gain schedule LPV output feedback control for IPMSM model presented in equations (4-10) adopting induced L_2/L_∞ -norm specifications in LPV framework. Generalized LPV open loop system with the following state space realization is considered:

$$G(\Theta) = \begin{cases} \dot{x} = A(\Theta(t))x + B_1(\Theta(t))w + B_2(\Theta(t))u \\ z = C_1(\Theta(t))x + D_{11}(\Theta(t))w + D_{12}(\Theta(t))u \\ y = C_2(\Theta(t))x + D_2(\Theta(t))w \end{cases} \quad (25)$$

where w is external input, and z is controlled output vector.

The state space representation of the gain schedule LPV controller is:

$$K(\Theta) = \begin{cases} \dot{x}_k = A_k(\Theta(t))x_k + B_k(\Theta(t))y \\ u = C_k(\Theta(t))x_k + D_k(\Theta(t))y \end{cases} \quad (26)$$

The LPV controller similar to the parameter dependence of the IPMSM ensures along all possible trajectories of $\Theta(t)$; the internal stability and minimization of the performance criterion from w to z output as follows:

$$\|T_{z,w}\| = \frac{\sup \|z\|_2}{\sup \|w\|_2} < \gamma \quad (27)$$

Theorem: Consider the given open loop LPV plant (25), with all possible $\Theta(t)$ trajectories, there exists a gain scheduled LPV output feedback controller (27) enforcing closed-loop asymptotic stability and an upper bound $\gamma > 0$ on the

L_2 -performance conditions of the closed loop system (27) from w to z , if there exists parameter dependent symmetric matrices X , Y quadruple of state space data $A_k(\Theta(t))$, $B_k(\Theta(t))$, $C_k(\Theta(t))$, $D_k(\Theta(t))$ such that the following LMI conditions holds:

$$\begin{bmatrix} AY + B_2\tilde{C}_k + * & * & * & * \\ \tilde{A}_k^T + A & A^T X + \tilde{B}_k C_2 + * & * & * \\ B_1 & XB_1 + \tilde{B}_k D_2 & \gamma I & * \\ C_1 Y + (D_{12}\tilde{C}_k^T) & C_1 & D_{11} & \gamma I \end{bmatrix} < 0 \quad (28)$$

$$\begin{bmatrix} X & I \\ I & Y \end{bmatrix} > 0 \quad (29)$$

After satisfying LMI conditions in equations (28) and (29) based on Bounded Real Lemma (BRL) [52] of parameter dependent decision matrices, $A_k(\Theta(t))$, $B_k(\Theta(t))$, $C_k(\Theta(t))$, $D_k(\Theta(t))$ and parameter dependent symmetric matrices X , Y , the vertex controllers can readily be computed by solving M and N from singular value decomposition:

$$I - XY = NM^T \quad (30)$$

where M and N matrices are invertible in case of full order controller.

$$\begin{cases} D_k = 0, \\ C_k = (\tilde{C}_k^T - D_k C_2 Y)(M_{-T}), \\ B_k = N^{-1}(\tilde{B}_k - XB_2 D_k), \\ A_k = (-N_{-1})(\tilde{A}_k - NB_k C_2 Y - XB_2 C_k M^T \\ -X(A + B_2 D_k C_2)Y)(M^T)^{-1} \end{cases} \quad (31)$$

The utilization of vertex property [52] of appropriate intervals space of scheduling parameters to convert the parameter dependent matrices in equation (17) to finite set of LMIs to be solved for unknown matrices and γ guaranteeing the L_2 gain performance.

Depending on the results described above, LPV controller is designed for the IPMSM based EV dynamics presented in section III. The design objectives are IPMSM stability over wide operating speed range, fast and accurate convergence in presence of load torque, and ability to reject noise.

To achieve these objectives, the design process of LPV controller is summarized as follows:

- A mixed H_∞ sensitivity loop shaping scheme is constructed as shown in Fig. (3). It includes complementary sensitivity W_S , sensitivity W_T , and control sensitivity W_K . Whereas, w is the desired input which is reference current and z is a vector $z = [z_s \ z_T \ z_K]^T$ in equation (25) that is the output of weighting matrices.
- Keeping in view the design objectives, the weights are tuned. The parameter values of weighted gains have been mentioned in appendix.
- The design specifications are expressed as an LMI problem in equations (28) and (29).
- LMI convex optimization approach and singular value decomposition is adopted in equations (30) and (31) to

TABLE 2. IPMSM parameters and EV specification data.

Parameters [units]	Symbol	Value
Stator Resistance[Ω]	R_s	0.5
Inductance in q-axis[H]	L_q	0.005
Inductance in d-axis[H]	L_d	0.0035
Flux Linkage[Wb]	ψ_{PM}	0.33
Pole pairs	p	3
Inertia[Kgm^2]	J	0.004
Viscous Damping	B	0.0028
Vehicle Data		
Gear ratio	G_r	6
Wheel radius[m]	w_r	0.3
Vehicle mass[kg]	m	750

solve the LMI problem and compute LPV controller gains.

C. ROBUST TORQUE AND FLUX CONTROLLER

Due to robustness and easy to implement properties [45], PID is reckoned for nonlinear torque and PM flux controller. It depends on the corrected coefficient values of Proportional, Integral and Derivative control terms illustrated in equation (32) and the parameters values have been mentioned in appendix. The proportional and integral terms usually decrease the rise time and steady state error. Mathematically, it can be represented as follows:

$$u(t) = K_p e(t) + K_i \int_0^t e(t)dt + K_d \frac{de(t)}{dt} \quad (32)$$

V. EV PERFORMANCE EVALUATION RESULTS

This section evaluate the improved torque performance of 3 kW IPMSM under extreme temperatures, part load and start/stop operation using LPV based FOC. The specifications of EV and 3 kW IPMSM are illustrated in Table 2. The proposed LPV observer-controller framework for EVs as shown in Fig. 1 is tested against different standard driving cycles adopted by automotive community [5], [53]. The driving cycles include periods of constant acceleration, deceleration, and various speed in urban and highway driving areas. The permanent magnet flux linkage reference is generated by solving equations (7-10) using equilibrium points while the reference torque is generated using the following equation:

$$\tau_L = \frac{F_{te} w_r}{G_r} \quad (33)$$

A. DRIVING CYCLES

For performance validation, EV simulator has been constructed in Matlab/Simulink using WLTP class 3, NEDC and FTP predefined driving cycles to generate the required torque. The IPMSM torque demand and operating speed over

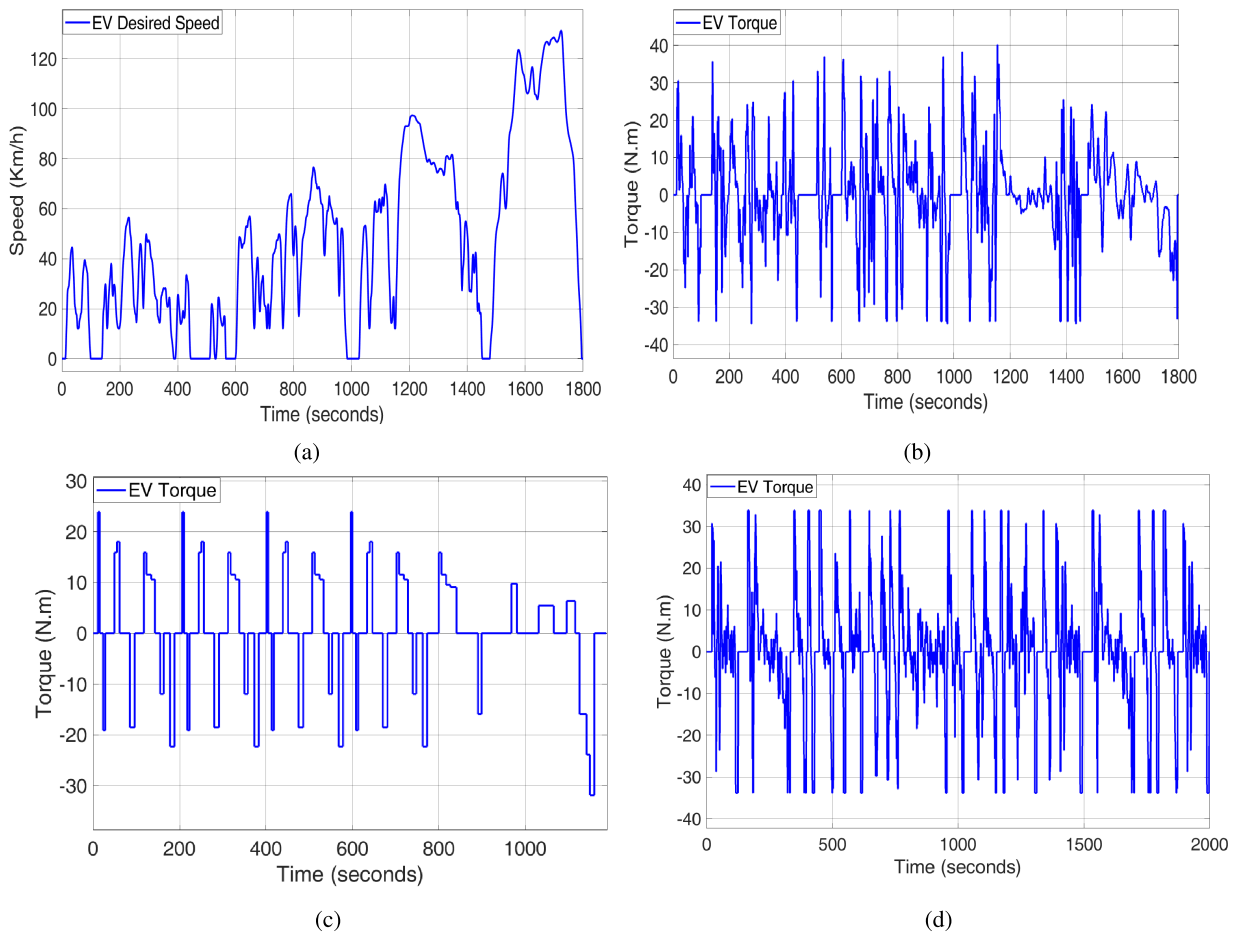


FIGURE 4. Simulation Results: Different driving cycles vehicle speed and torque required (a) The WLTP class 3 driving cycle vehicle speed (b) The WLTP class 3 driving cycle illustrates the required torque for EV (c) The NEDC driving cycle illustrates the torque load required for EV (d) The FTP driving cycle illustrates required torque for EV.

driving cycles are specified via vehicle mechanics mentioned in equations (1)–(3) and using the specifications mentioned in Table 2 can be seen in Fig. (4(a), (b), (c) & (d)). It is worth mentioning that the state-of-the-art control architecture designed in section IV can be used for any driving cycle.

B. EVALUATION OF CONTROL ARCHITECTURE

In order to evaluate the performance of proposed LPV controller with observer strategy for managing the thermally degrading of torque for EV powertrain, a dynamic stator resistor profile based on equation (14) is generated as shown in Fig. (5). The profile at different temperatures 20°C , 35°C , 50°C and 65°C for the complete driving cycles is simulated to demonstrate the effectiveness of control architecture depicted in Fig. (1).

To gauge the performance of IPMSM, the most substantial element of electrified powertrain, the desired speed and torque need to be evaluated based on equations (18 and 19) for all temperature environments. It can be clearly seen from Fig. (1), in the presence of prompt changes in torque from driving cycles, the estimation of degraded torque due to

temperature is ensured through LPV observer designed in section III-A.

Figs. (6) and (7) validate the LPV observer-controller robust tracking performance of EV speed against the WLTP class 3 and NEDC driving cycle for temperature profiles at 35°C and 20°C respectively. It can also be seen that the proposed control strategy performs efficient tracking at very low speed near zero and high speed near 120 km/h generated by real time driving conditions considering WLTP class 3 and NEDC driving cycles.

Fig. (8) shows fast convergence for compensating of torque with respect to change in temperature at 35°C for entire period of WLTP class 3 driving cycle in Fig. (4b) from as low as -35 Nm to as high as 42 Nm . A portion from 1120s to 1140s of the requested PM flux linkage and estimated flux linkage has also been illustrated in Fig. (12) to show effectiveness of the estimation used in the compensation of thermally derated torque depicted in Fig. (8). In Figs. (10) and (11), flux errors have also been demonstrated. Fig. (11) depicts fast convergence achieved with in 0.10s, and convergence error can be seen near zero at 715.45s against FTP driving cycle. The simulation results for torque compensation

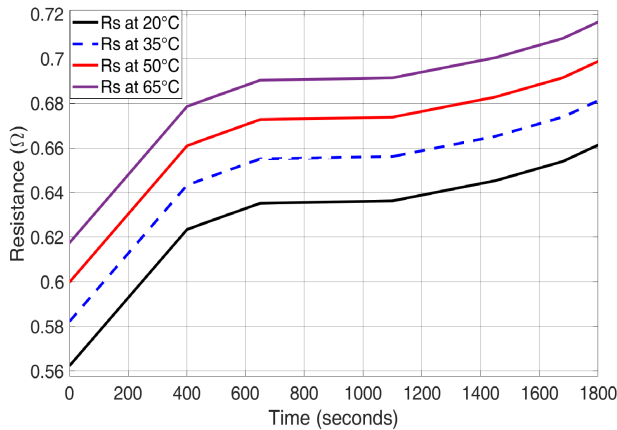


FIGURE 5. Simulation Results: Change of stator resistance w.r.t to temperature at entire operation period.

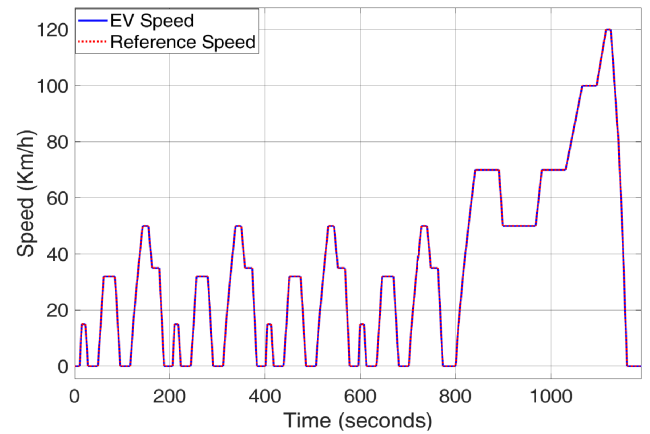


FIGURE 7. Simulation Results: EV speed in (Km/h) against NEDC at 20°C.

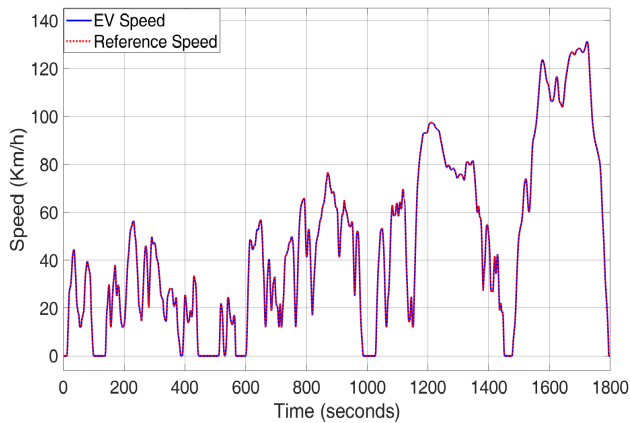


FIGURE 6. Simulation Results: EV speed in (Km/h) against WLTP class 3 at 35°C.

TABLE 3. Performance indices of LPV based FOC torque tracking at various ambient temperature.

Ambient Temperature	Mean Square Error		
	WLTP class 3	NEDC	FTP
20°C	0.259	0.385	0.429
35°C	0.260	0.385	0.430
50°C	0.286	0.388	0.483
65°C	0.289	0.401	0.477

validates the fast convergence and robustness of the control architecture.

In addition to that Table 3 and 4 present LPV control technique performance indices of Mean Square Error (MSE) for torque and speed tracking against various ambient temperature profiles for WLTP, NEDC and FTP driving cycles.

Due to variation in acceleration, deacceleration, torque and speed in urban and highways, the MSE error ranges from 0.259 to 0.289 for torque and 0.031 to 0.035 for speed against

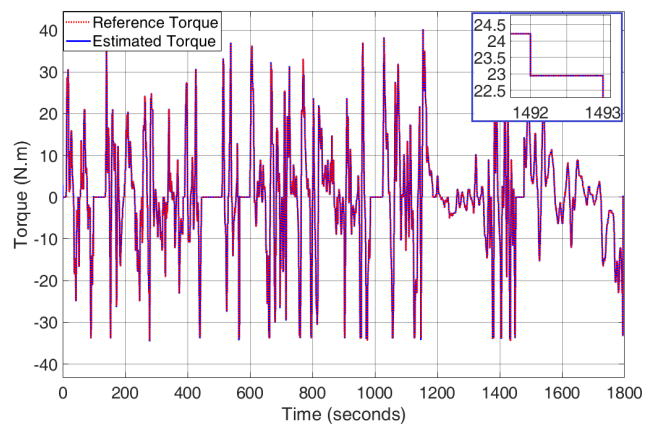


FIGURE 8. Simulation Results: Compensation of Torque w.r.t to change in temperature (35°C) at entire operation period for WLTP class 3 driving cycle.

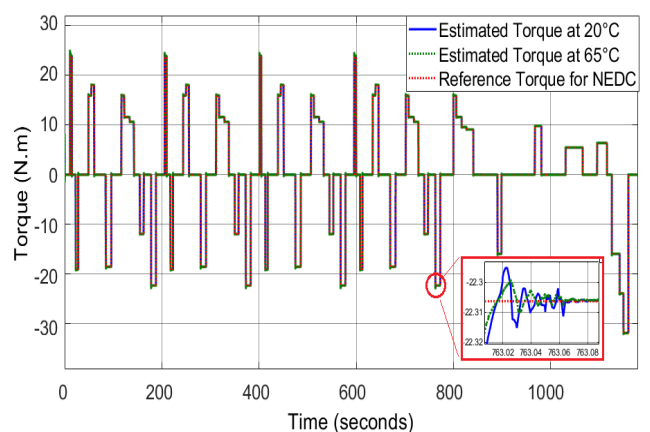


FIGURE 9. Simulation Results: Compensation of Torque w.r.t to change in temperature (20°C and 65°C) at entire operation period for NEDC Driving cycle.

the WLTP driving cycles considering the various of ambient temperature. The highest MSE error of torque is 0.477 at temperature of 65°C against FTP testing. For NEDC driving

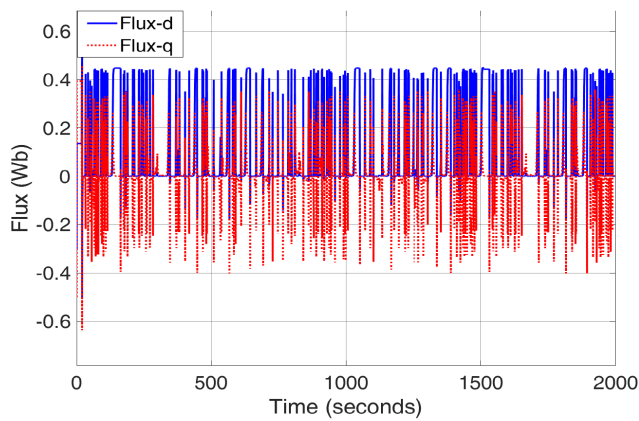


FIGURE 10. Simulation Results: Flux estimation error for complete operating range.

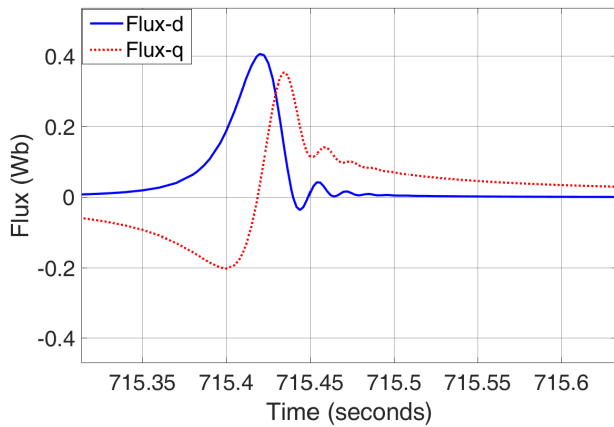


FIGURE 11. Simulation Results: Zommed version of flux estimation error.

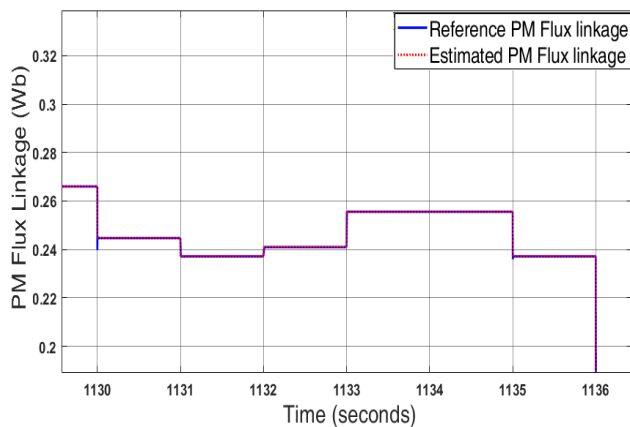


FIGURE 12. Simulation Results: PM flux estimation and requested PM flux against the reference torque at 35°C.

cycle the MSE of speed varies from 0.385 to 0.401. Hence, the robustness and efficacy of FEV speed tracking and compensation strategy of derating torque due to parameter variation against change of temperature is validated.

TABLE 4. Performance indices of LPV based FOC speed tracking at various ambient temperature.

Ambient Temperature	Mean Square Error		
	WLTP class 3	NEDC	FTP
20°C	0.031	0.023	0.021
35°C	0.033	0.023	0.022
50°C	0.033	0.024	0.021
65°C	0.035	0.026	0.023

TABLE 5. Average error in percentage of EV Torque and speed.

Performance	Average Error (%)		
	WLTP class 3	NEDC	FTP
Torque	5.5	1.2	2.5
Speed	5.9	0.1	3.5

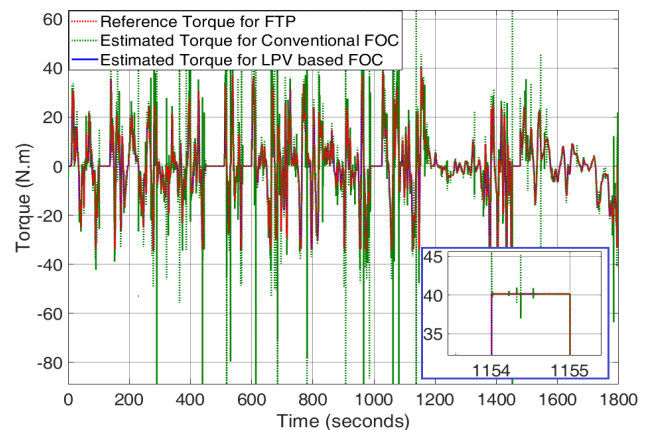


FIGURE 13. Simulation Results: Comparison of LPV based FOC and conventional FOC against WLTP class 3 driving cycle.

The performance in terms of percentage error is calculated against the norms mentioned in equations (18) and (19) mentioned in Table 5. Hence, the robustness and efficacy of EV speed tracking and compensation strategy of derating torque due to parameter variation against change of temperature is evaluated.

VI. COMPARISON WITH CONVENTIONAL FOC

The control architecture LPV based FOC controller-observer pair designed in section IV is compared with conventional FOC. The experiments simulations are performed for all driving cycles at ambient temperature. The control structure, LPV based FOC results outperform the conventional FOC against WLTP, NEDC and FTP driving cycles as shown in Figs. (13), (14) and (15). The graphs clearly illustrate a very high setting and rise time of conventional FOC as compared with the LPV based FOC. The LPV based FOC outperformed conventional

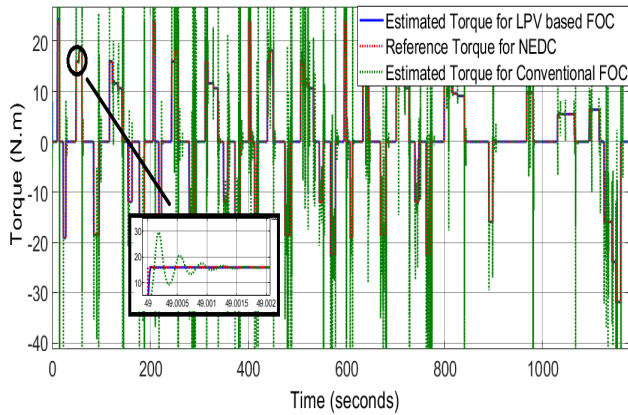


FIGURE 14. Simulation Results: Comparison of LPV based FOC and conventional FOC against NEDC driving cycle.

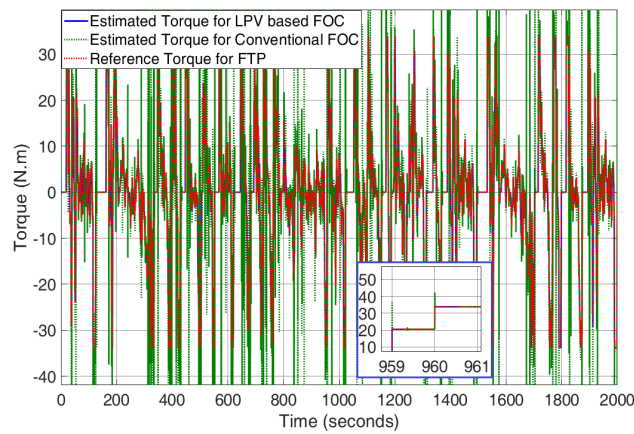


FIGURE 15. Simulation Results: Comparison of LPV based FOC and conventional FOC against FTP driving cycle.

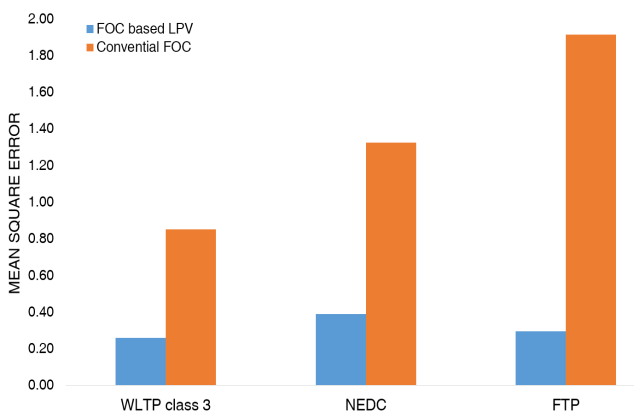


FIGURE 16. MSE Torque Comparison of LPV based FOC and Conventional FOC.

with the error difference of 0.70. For comparative analysis of the control techniques, mean square error for torque tracking has been computed as a performance indice. Fig. (16) shows an exponential low MSE 0.259 with LPV based FOC against

WLTP driving cycle as compared with conventional FOC to be 0.84, which is more than 200 % increase in error. Furthermore, when compared to the LPV based FOC, the experiments with NEDC and FTP for conventional FOC show even higher percentages of error.

VII. CONCLUSION

A novel approach based on LPV to address derating of torque in EV is designed and evaluated. The robust performance of the observer-controller for managing the thermal effects has been corroborated. The efficiency of IPMSM based EV has been validated using WLTP class 3, NEDC and FTP driving cycles through extensive experiments simulations in Matlab/Simulink. The experiments simulation results established fast convergence and stability of LPV based FOC for flux estimation and current controller respectively. The effectiveness of LPV based FOC with change in ambient temperature has been demonstrated for compensating the derating of torque for wide operating conditions. The proposed control strategy outperforms in comparison to conventional FOC against the same conditions. The experiments simulation results, MSE value, percentage error and comparative analysis established the efficacy and robustness of the proposed scheme. In future, saturation and harmonics along with thermal effects will be explored, analysed and compensated for IPMSM based EVs.

APPENDIX

The weight functions of LPV current controller are tuned iteratively in interest of the following:

- Robust stability with tracking error less than 1% and response time of 5 seconds

$$W_s = \begin{bmatrix} \frac{0.1s + 800}{s + 7.2} & 0 \\ 0 & \frac{0.1s + 800}{s + 7.2} \end{bmatrix}$$

- Achieve overshoot of the system less than 11%

$$W_T = \begin{bmatrix} 0.9 & 0 \\ 0 & 0.9 \end{bmatrix}$$

- To limitize control action to 60dB

$$W_K = \begin{bmatrix} 0.0001 & 0 \\ 0 & 0.0001 \end{bmatrix}$$

The computed values of PID controller for torque and flux controller are $k_p = 0.39$, $k_i = 76720$, $k_d = 1.17$.

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